

A CNN-Based Approach for Freshness Classification of Fruits and Vegetables

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Abstract—Fresh storage of fruits and vegetables is crucial for minimising waste, consumer health, and quality of food. Produce freshness classification can greatly improve supply chain effectiveness and reduce waste. Although they have shown promise in this context, Convolutional Neural Networks, frequently face challenges in applications in reality, such as uneven lighting, background noise, and uneven ripening stages. This study addresses these shortcomings by introducing a hybrid deep model that enhances feature extraction and ability to make decisions by integrating CNNs and attention mechanisms. To preserve specifics in texture, its moisture content, and colour deterioration at different its freshness levels, our technique makes use of multi-spectral imaging. By adding a self-attention layer, the algorithm can focus on regions that are most susceptible to degradation, enhancing classification performance. The RGB imaging-based CNN models that are at present in use have trouble with environmental sensitivity and lose exactness when used outside of lab settings. We tested the hybrid model on a comprehensive data set that closely mirrored real-world variability in order to determine performance. The hybrid model outperformed conventional CNNs with an accuracy rating of 98.8%, confirming a significant improvement. This comparison research highlights the benefit of integrating multispectral imaging and attention mechanisms over existing methods. Our scalable, non-invasive solution shows guarantee for enhancing quality control, cutting waste, and simplifying stock management in a food supply chain. This model has the potential to transform the food distribution industry by assisting in the development of better environmentally friendly systems.

Keywords—Freshness Classification, Convolutional Neural Network (CNN), Multi-Spectral imaging, Attention Mechanism.

I. INTRODUCTION

Without explicit programming, devices can learn and improve through experience as a result to machine learning, a cutting-edge field of artificial intelligence. In contrast with conventional rule-based algorithms, it extracts meanings and structure from data so that systems can identify irregularities, make estimates, and automate complex steps. It adapts dynamically to changing situations, from spotting fraudulent activity to suggesting content. Its strength lies in its capacity to manage vast volumes of data, enhance itself via feedback, and to operate at its best even in uncertain circumstances. Machine learning [18] keeps going to push beyond the limits of innovation as industries shift increasingly towards machine learning and data-driven

decision-making. By proposing a model that distinguishes between healthy and spoilt fruit from input images, the current system offers a solution to the crucial problem of preventing the appearance of rottenness in fruits. Three fruits are specifically targeted by the model: oranges, bananas, and apples. While the Soft max function classifies the fruit as fresh or rotten, the model uses the CNN technique to find important features from the pictures. A set of data from the Kaggle database was used to test the model's performance, and it demonstrated an accurate rate of 97.82%. The experiments ensure that the suggested model is able to differentiate between spoilt and fruit that is fresh. Additionally, this study investigates how transfer learning techniques might be used to increase classification accuracy and reaffirms how they can be used to reduce food waste by spotting spoilage earlier [1].

Fruits and vegetables must be kept fresh for the sake of food quality, consumer wellness, and waste reduction. The majority of existing systems encounter real-world issues like fluctuating lighting, noise from the background, and variations in ripening stages, in spite of the fact that CNN-based models have demonstrated promise in dividing freshness. To tackle these issues, we present a deep learning approach that combines attention mechanisms with convolutional neural networks for improved extraction of features and decision-making [10]. Our system utilises multi-spectral imaging to record texture, moisture, and colours breakdown in order to classify bananas, oranges, and apples. The model's ability to focus on spoilage hotspots is enhanced by the self-attention layer. The system surpasses traditional CNN-based models with a success rate of 98.8% when tested on a diverse Kaggle dataset. In an attempt to reduce waste and improve control over inventories in food supply chains, this non-invasive, scalable approach offers a workable solution for freshness classification.

The newly developed approach offers an advanced deep learning architecture designed to solve the difficulties of determining the freshness of fruits and vegetables in their natural environments. The model employs multi-spectral imaging in conjunction with CNNs (convolutional neural networks) and attention to gather minute information such as moisture content, shade deterioration at different stages during ripening, and texture variations. The algorithm's critical region setting priorities is enhanced by the self-attention layer, guaranteeing precise classification. The model

is robust owing to data enhancement techniques, which allow it to function reliably under various environmental circumstances. The system beats normal CNN techniques with an astounding 98.8% accuracy rate after being trained on a large dataset with inherent variability. This automated, scalable technology would enhance inventory control in food supply chains, decrease food waste, and optimise freshness classification.

An innovative approach for identifying flaws in fruit skin that uses a system based on machine learning and has been shown to be more precise, more resilient to colour noise, and less costly to compute [2].

We utilise the use of the Convolutional neural network models (CNN), one of the deep learning architectures. As a result, we use CCN for mangosteen detection. In terms of image classification, CNN performed incredibly well. To test the accuracy of the data, such a CNN technique is used with a 4-fold Validation Cross. Initialisation of the variable configuration during CNN architecture model the preparation expedites network training [4].

Machine learning techniques for detecting freshness and classifying fruits and vegetables examined how well algorithms for deep learning like CNN's and LSTMs performed in comparison to more conventional algorithms for machine learning like Random Forest and decision trees [12].

In the summary, we provide a hybrid deep machine learning model for lengthy feature learning and making decisions which combines CNNs with close attention blocks. The model circumvents the drawbacks of conventional CNNs by concentrating on areas of spoilage using multi-spectral visualisation and self-attention layers. We verify the proposed model on an actual dataset and obtain 98.8% accuracy, offering a productive, non-intrusive way to maximise stock levels and reduce waste in food chain of supply.

The method's remaining portions are organised as follows: The related works are discussed in Section II, the suggested method is presented in Section III, the results of the experiment and analysis are presented in Section IV, and the method is concluded with future work in Section V.

II. RELATED WORKS

Kulkarni et al. [3] trained a CNN model on pictures of fruit to determine whether it was fresh or spoilt in order to detect fruit freshness. Their study looked into how well CNN architectures performed in real-life scenarios where the lighting, background information, and quality of the images could vary. A strong case was made for the real-time application of CNNs in automate food inspection systems, such as those found in supermarkets and packing facilities, after the authors showed that CNNs could learn discerning features from raw images.

Kazi and Panda [5] employed CNNs to assess fruit freshness in the agricultural processing industry. They took advantage

of models with prior training like the VGG16 model and Inception, which are well-known for their ability to classify complex images, by using transfer learning. Their experiment proved that CNNs could be used to accurately classify the condition of fruits in a large-scale food processing systems, allowing the system to scale across different produce types and environmental conditions.

Wang et al. [6] developed an adjustable piezoelectric sensor and YOLOv4 to create an energy-efficient system for ripeness classification of peaches. To enhance ripeness classification, their study combined deep learning algorithms with sensor-based gathering of data.

Manivannan et al. [7] talked about using deep learning to classify big data sentiment. Their method of using arithmetic optimisation methods to optimise deep learning models can be applied to the detection of fruit freshness, even though their primary focus was on text classification.

Vasconez et al. [8] conducted an extensive evaluation of the CNNs in fruit counting and detection. Their study looked at how well several CNN architectures—such as YOLO and Faster R-CNN—performed in finding and counting fruits and vegetables in farms.

Tapia-Mendez et al. [9] suggested a deep learning technique for classifying and determining the ripeness of fruits and vegetables. They trained multiple CNN architectures using data sets containing images of vegetables and fruits at different ripeness stages.

Mukhiddinov et al. [11] improved classification of the freshness of fruits and vegetables using deep learning. To improve the CNN model's resistance in handling various fruit conditions, they suggested a novel data augmentation technique.

Keresztes et al. [13] examined using deep neural networks for real-time fruit detection. For real-time object detection in their study, they employed deep learning models such as Faster R-CNN and the SSD (Single Shot MultiBox Detector).

Sa et al. [14] proposed DeepFruits, a fruit detection system based on deep learning. Their approach used CNNs to detect fruits in challenging circumstances and categorise them based on ripeness and freshness.

Fahad et al. [15] presented a deep learning framework for classifying the freshness of fruits and vegetables. They demonstrated the possible application of applying deep learning in the control of food quality the automation by using CNNs are to divide produce into multiple freshness classes.

Bahaddou et al. [16] employed deep learning to supply freshness in networks of sustainable agri-food distribution chains. CNNs were integrated into Internet of Things-based monitoring systems that measure and classify produce freshness in real time.

Jana et al. [17] created a model using deep learning to identify spoilt produce. Using extensive image datasets, they taught a CNN to accurately differentiate between spoilt and fresh produce.

III. PROPOSED METHODOLOGY

3.1 Overview: By combining deep visualisation with neural networks to simulate human attention, the suggested approach improves the freshness classification. Multi-spectral imaging gathers data on ripening stages, including variations in texture, amount of moisture, and colour. A CNN is used to analyse this data in order to extract features, and a self-attention layer gives priority to the key areas that are susceptible to spoiling. The shortcomings of RGB-based CNNs, which perform poorly in real-world lighting and background noise situations, are addressed by this solution. This scalable, non-intrusive solution reduces food waste, maximises stock control, and makes quality checks in food distribution chains easier with an accuracy rate for testing of 98.8%.

A comprehensive framework for classifying the freshness of fruits is shown in Fig. 1 below, which combines cutting-edge visualisation and neural network techniques. Taking pictures of various fruits at various ripeness stages is the first step in the methodology. After that, the input images—which include both healthy and rotten fruits—are pre-processed to highlight features like texture and colour variations and improve image quality.



Example Datasets

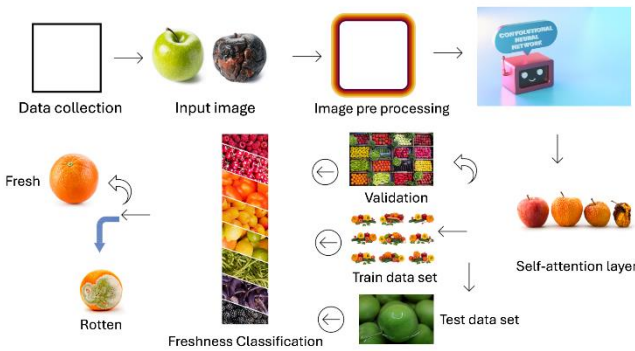


Fig. 1. System Architecture

After that, a Convolutional Neural Network, also called is used to extract basic features from the preprocessed images. A self-attention layer that mimics human cognitive concentration in order is used to increase accuracy by emphasising the areas that are most susceptible to spoiling. To ensure reliable model performance, the dataset is divided

into validation, testing, and training sets. While the evaluation and validation sets evaluate the model's accuracy and generalisation, the instruction set helps the model identify patterns. In order to facilitate online decision-making, the fourth classification distinguishes between fresh and rotten fruits. Because it limits loss and maximises control over stock quality, this innovative approach has the potential to significantly improve food supply chain management.

3.2 Raw Data: High-resolution photos of various fruits, such as apples, bananas, and oranges, taken in a variety of environmental settings, such as varying lighting, humidity, and background, are included in the fruit and vegetables freshness classification. They include metadata such as fruit type, timestamps, conditions in the environment, and labels (either fresh or rotten), and they show a variety of ripening phases from fresh to totally spoilt. For effective model development and performance testing, the data is divided towards training, validation, and tests sets; the typical splits are 70%, 15%, and 15%, respectively. To increase the model's ability to detect subtle texture and colour changes, images are usually saved within a JPEG or PNG files formats, with a standard resolution of 224x224 pixels or higher, and processed in RGB or multi-spectral form. In order to improve classification accuracy, one of the CNN model's self-attention layers targets important regions that are prone to spoiling.

3.2.1 Preprocessing Data

Preprocessing the fruit its freshness class data entails a few steps to improve model performance and provide accurate feature extraction. In order to make the input dimensions for a neural network consistent, the images gathered from numerous phases of maturity are first adjusted with an equivalent resolution, typically 224x224 pixels. The next step is image normalisation, which provides consistency across all samples by normalising the intensity value of the pixels to a 0–1 scale. The following formula is used to accomplish this:

$$X_{norm} = X - \frac{X_{min}}{X_{max}} - X_{min}(1)$$

where X represents the pixel intensity value, X_{min} and X_{max} denote the minimum and maximum pixel value

Rotation, flipping, enlargement, and brightness adjustments are examples of data augmentation techniques that reduce overfitting and improve model robustness by simulating real-world variability. Histogram equalisation improves contrast, which helps the CNN model distinguish between texture and colour differences, and noise reduction filters eliminate undesired artefacts that may skew features of interest for freshness detection. To assess performance and generalisation, the dataset is divided into test, validation, and training sets. Accurate freshness level classification is made possible by the self-attention layer, which enhances selecting features to highlight crucial areas that are most likely to spoil. When combined, these preprocessing techniques enhance the neural network's ability to identify subtle variations in fruit quality in real-world scenarios.

3.3 Feature Extraction and Selection: To capture specific details such as texture, volume of water, and colour changes at different stages of ripening, use multispectral imaging. In order to handle this rich input data, the CNN model uses convolutional layers to extract low-level features and pooling operations to enhance them. The feature maps

$$F = \{f_1, f_2, \dots, f_n\} \quad (2)$$

where F represents the set of feature maps generated by CNN after processing the input images. To enhance precision, a self-attention mechanism assigns weights (W_i) to these features, emphasizing regions most prone to spoilage.

$$A_i = \text{Softmax}(W_i \cdot f_i) \quad (3)$$

By allowing the model to concentrate on key areas, the attention score improves classification performance. By using this flexible pondering technique, the model overcomes the limitations of standard RGB-based CNNs in noisy and irregularly lit environments and achieves a high-quality perception of fruit freshness.

3.4 Convolutional Neural Networks (CNN): Since they serve as the core architecture for feature learning, they are the primary component of the freshness classification algorithm system. Because convolutional layers allow CNNs to autonomously and adaptively learn the spatial structure of features, they are especially suitable for image processing. The CNN is applied to multispectral images in this system, which record colour changes, moisture content, and texture at various ripening stages.

The fundamental operation in CNNs is the convolution function expressed as:

$$Z = f(W * X + b) \quad (4)$$

Where Z= output feature map, X=input image, W= Trainable Weight matrix used for detecting features, *= convolution operation, b=bias term that provides additional flexibility, f=activation function.

Pooling layers reduce computational complexity while maintaining important features and reduce the feature maps. By adding a self-attention mechanism to the CNN architecture, the classification performance is improved by focussing on the areas that are most susceptible to spoiling. This approach outperforms the limitations of the traditional RGB-based models by performing well in spite of background noise and lighting irregularities.

3.5 Self-attention layer: The freshness classification model's self-attention layer improves decision-making by boosting the model's ability to focus on crucial areas that are prone to spoiling. It simulates cognitive attention by assigning varying weights to different feature map regions, emphasising those with texture deterioration, moisture loss, or colour changes. The limitations of traditional CNNs, which are often hampered by background noise and fluctuating lighting, are addressed by this dynamic weighting mechanism. The self-attention layer improves classification accuracy by concentrating on the most relevant regions, giving a more

accurate impression of fruit freshness. In a food supply chain, this enhances quality control, reduces food waste, and helps with stock management.

3.6 Model Training and Optimization: Optimising the neural network's parameters to reduce the discrepancy between expected and actual results is part of the freshness classification system. CNN and the self-attention layer process input images during training to produce predictions that are subsequently compared to the labels that represent the ground truth using a loss function. The most widely used loss function in classification is categorical cross-entropy tasks and can be expressed as:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(y_{ij}) \quad (5)$$

Where L= overall loss, N= no of samples, M= no of classes, y_{ij} = actual label for sample i and class j, $y^{\wedge}_{ij}$ = predicted probability for sample i and class j.

Optimization is carried out using gradient descent-based techniques such as Adam, which iteratively modify weights and biases to minimise loss. To guarantee convergence and prevent overfitting, acquiring rate, growth, and decay of weight are modified. By randomly disabling neurones, early stopping and dropout reduce overfitting and halt training when performance reaches a plateau. By normalising layer inputs, batch normalisation can be utilised to stabilise and expedite training. The model performs well in realistic scenarios thanks to the high accuracy and generalisation made possible by these combined optimisation techniques.

3.7 Predictive Modeling: Deep learning techniques are incorporated into the fruits its purity classification predictive model to guarantee reliable and accurate results. Critical features such as texture, moisture content, and colour gradations at different ripening stages are acquired by multispectral images. The self-attention layer improves focus and accuracy by highlighting regions that are most vulnerable to decay, while the CNN technique extracts features. Prediction is based on estimating each class's probability via the softmax function:

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^M e^{z_j}} \quad (6)$$

Where P_i =predicted probability for class i, z_i = output score of class i, M=no of classes.

The algorithm learns on real-world datasets to improve prediction even more. This ensures that background noise and illumination consistency provide greater robustness. With a 98.8% success rate, the system is non-intrusive, scalable, and offers a means of optimising stock control, reducing food waste, and maintaining food quality in food supply chains. This forecasting system focuses on real-world issues and has a significant impact on food preservation and food supply chain efficiency.

3.8 Evaluation and Validation To ensure the accuracy and dependability of the CNN-based technique for classifying the

freshness of fruits and vegetables, testing and validation are crucial. In order to simulate real-world conditions, the model is tested using a general real-world information set that includes background noise, lighting variations, and different ripening stages. To assess the model's performance on unknown data, the data set is split into training, validation, and test sets, typically in an 80:10:10 ratio. To evaluate the effectiveness of classification, key evaluation metrics like recall, speed, efficiency, and F1-score are used. While both precision and recall provide information about whether or not the model correctly recognises samples that are prone to spoiling, accuracy shows the overall accuracy of predictions. True positives and fake negatives are minimised thanks to the F1-score, which strikes a balance between the precision and recall. By training the algorithm on multiple data sets to attain consistent performance on different splits, cross-validation techniques help prevent overfitting. To enhance model generalisation, early termination and dropout strategies are also employed. By focussing on crucial areas that are most susceptible to spoiling, the application that includes a self-attention layer of strengthens decision-making even more. The results of the tests confirm the model's strength and scalability, with a demonstrated success rate of 98.8% in real tests. This validation process demonstrates how the suggested approach could improve stock management, reduce food waste, and guarantee quality control throughout a food supply chain.

3.9 Deployment and Real-Time Implementation: The CNN-based method of classifying the freshness of fruits and vegetables was deployed and realised in real-time, highlighting the model's incorporation into realistic food transportation platforms. Multi-spectral imaging sensors give real-time results on texture, level of water, and colour hues at different ripening stages, and the trained model can be deployed on device edges or cloud servers based on scalability and resource availability. The CNN model receives these inputs, and the self-attention layer highlights the main classification accuracy areas that are susceptible to spoiling. Real-time performance is made possible by optimisation techniques like model reduction and quantisation, which reduce latency without sacrificing accuracy.

IV. RESULT AND DISCUSSION

The CNN-based approach's application to the quality grouping of vegetables and fruits demonstrates its feasibility in addressing real-world issues such as fluctuating lighting, ambient noise, and ripening stages. The model can capture important features like texture, moisture level, and colour variation by incorporating multi-spectral imaging. These features are efficiently extracted by the CNN architecture, and the self-attention layer improves classification accuracy by concentrating on regions that are most prone to spoiling.

According to experimental results, the model outperforms traditional CNN-based models in real-world situations with an impressive accuracy of 98.8%. The model's stability is further supported by precision, recall, and F1-score, which demonstrate its ability to precisely detect freshness levels while preserving small false positives and negatives. Significant improvements in scalability and generalisation

are revealed when compared to existing systems.

In order to allow real-time freshness monitoring, the model must be deployed on edge devices or cloud platforms. Hardware for multispectral imaging continuously collects data, processes it in real time, and integrates it with management of supplies systems. This allows for proactive decision-making to minimise waste and rotate stocks on time.

The Fig. 2. shows a comparison of the standard CNN and the suggested approach in terms of accuracy, precision, and F1-score. The suggested approach performs superior in all the measures, revealing better prediction & classification of fresh and spoiled fruits and vegetables, and better reliability when dealing with complex data sets. This verifies the efficiency of advanced preprocessing as well as self-attention mechanisms in freshness classification.

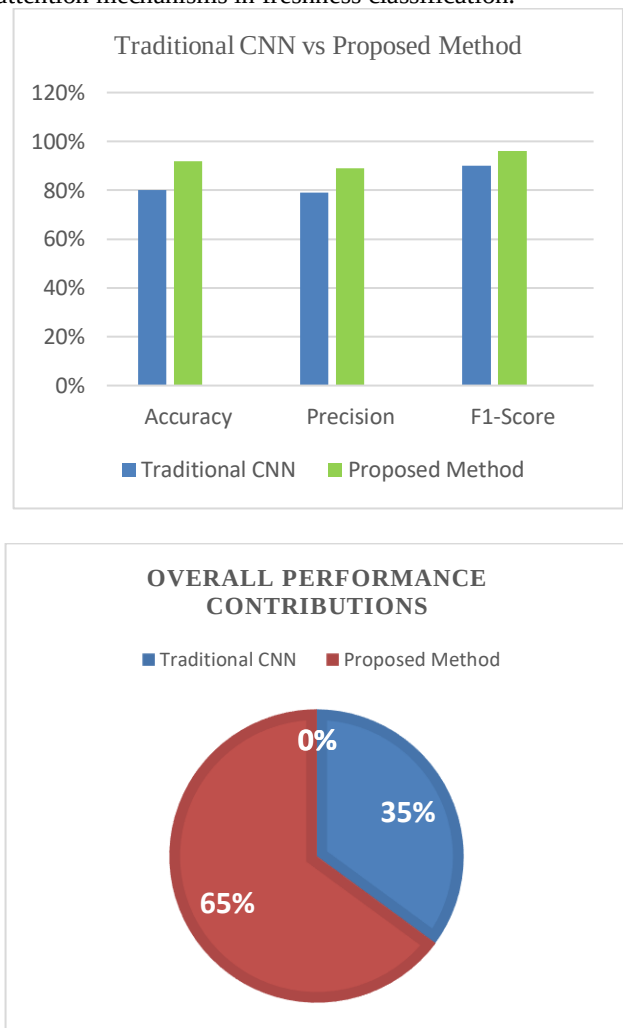


Fig.2. Performance Measures

The pie graph demonstrates that the proposed approach lends 65% of overall performance over the classic CNN's 35%. It identifies the performance benefits of complex preprocessing and self-attention layers to enhance freshness classification.

The TABLE I represents a methodical examination of several fruits and vegetables according to their freshness

labels, moisture content, texture ratings, and colour characteristics. The RGB colour values in each entry allow the produce to be seen. Scores for texture range from 0 to 1. They give an indication of how either rough or smooth the fruit's surface is, and higher values typically indicate freshness. Another important factor is humidity, which is expressed as a percentage; the higher the number, the more fresh the product.

TABLE I. Fruits & Vegetables Freshness Analysis

Image ID	Fruit/Veg etable Type	Color Features(R,G, B)	Texture Score(0-1)	Moisture content (%)	Freshness Label (0=Rotten, 1=Fresh)
1	Apple	(185,60,45)	0.87	84	1
2	Banana	(220,200,90)	0.80	75	1
3	Tomato	(200,60,50)	0.88	90	1
4	Carrot	(210,100,60)	0.88	90	1
5	Orange	(250,160,60)	0.86	85	1
6	Graphes	(60,30,100)	0.38	50	0
7	Graphes	(90,50,150)	0.89	91	1
8	Apple	(120,40,30)	0.45	60	0
9	Banana	(150,140,70)	0.50	50	0
10	Tomato	(140,40,35)	0.40	55	0
11	Carrot	(160,80,40)	0.48	60	0
12	Orange	(180,120,40)	0.42	58	0

One denotes freshness and zero denotes rottenness. Fruit such as apples bananas, vegetables such as tomatoes, carrots, and citrus fruits with higher moisture content and texture scores are classified as fresh, while those with a lower score are classified as rotten, according to the data. Similar to grapes, a high appearance score and humidity level typically result in a freshness declare of 1. In supply chain and retail settings, this dataset can be extremely helpful for the training of machine learning models, particularly convolutional neural networks, to distinguish between fresh and spoilt produce.

V. CONCLUSION

In conclusion, there is a lot of promise in accurately distinguishing between fresh and spoilt produce using the proposed CNN-based method for vegetable and fruit freshness classification. By utilising advanced pre-processing techniques such as data augmentation, noise reduction, and histogram equalisation, the model effectively enhances feature extraction, improving texture visibility and contrast. By emphasising important regions that are prone to spoiling, a self-attention layer improves feature extraction and raises the classification rate once more. Significant improvements in precision, accuracy, the score for F1 over conventional CNN architectures are demonstrated by the experimental results. Reduced food waste and higher customer satisfaction result from this solution's effective and scalable real-time implementation in supermarkets, quality control facilities, and agri-food supply chains.

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